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Shareholder Litigation and Corporate Innovation

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Abstract. We investigate whether and to what extent shareholder litigation shapes corporate innovation by examining the staggered adoption of universal demand laws in 23 states from 1989 to 2005. These laws impose obstacles against shareholders filing derivative lawsuits, thereby significantly reducing managers' litigation risk. Using a difference-in-differences design and a matched sample, we find that, following the passage of the laws, firms invested more in research and development, produced more patents in new technological classes and more patents based on new knowledge, generated more patents with significant impacts, and achieved higher patent value. Our findings suggest that the external pressure imposed by shareholder litigation discourages managers from engaging in explorative innovation activities.

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Keywords: shareholder litigation • innovation • patents • derivative lawsuit

1. Introduction

To what extent does shareholder litigation affect a firm's innovation activities? Research in finance so far provides little evidence to this question. Seminal research in law and finance (La Porta et al. 1997, 1998) suggests that shareholder litigation helps resolve agency problems arising from the separation of ownership and control. When officers and directors breach their fiduciary duties and abuse the power of their positions, shareholders are entitled to file legal claims against the wrongdoers. However, scholars have also raised the concern that many shareholder lawsuits are frivolous and waste firms' assets (Romano 1991). The burden imposed by shareholder litigation on managers reduces their incentives to experiment with new ideas (Kinney 1994). Thus, managers may consider excessive shareholder litigation as an "uncontrolled tax on innovation."¹

We explore the impact of shareholder litigation on managerial incentives in the innovation decisions by exploiting a staggered law change that reduces managers' exposure to shareholder lawsuits.² We test two conflicting hypotheses from the literature. First, the "disciplining hypothesis" argues that the threat of shareholder litigation "disciplines" a manager's behavior and stimulates corporate innovation. Without proper oversight, the managerial agency problems, on the one hand, may induce shirking or self-dealing behaviors (Jensen and Meckling 1976, Jensen 1986) and give rise to the career concern that discourages

risk-taking activities (Holmstrom 1999) on the other hand. The threat of shareholder litigation can mitigate such "moral hazard" issues and ensure that managers focus on innovation activities. Conversely, managers may abandon efforts in innovation search when the exposure to shareholder litigation is reduced.

Second, the "pressure hypothesis" suggests that shareholder litigation stifles corporate innovation. As opposed to the potential disciplining benefits, shareholder litigation may impose excessive short-term pressures on the management team and intensify the capital market concerns. The process of innovation involves the possibility of project failure and inadequate economic outcomes (Holmstrom 1989).³ Innovation failures will quickly translate into a decline in stock prices,⁴ which could be used as evidence for shareholder losses in the courts. Shareholders cannot fully observe the efforts in the innovative process, and thus, they may attribute the negative performance to a breach of fiduciary duty and file the shareholder suit. The potential litigation risk may undermine managerial incentives for innovation by restricting the tolerance of failure, which is perceived to be critical to motivate innovation in the prior literature (Azoulay et al. 2011, Manso 2011, Tian and Wang 2011). For example, Tesla Motors has transformed the landscape of the auto industry through its innovations for electric vehicles, such as battery and charging technology. However, in 2013, multiple battery fires on Tesla Model S sent Tesla's stock tumbling.

Those product failures led to a derivative lawsuit filed against Tesla's managers, alleging that they breached their fiduciary duties and materially damaged the company as evidenced by stock price drops.⁵ As asserted by senators in the U.S. Congress, "companies, particularly growth firms, say they are sued whenever their stock drops" (Seligman 1994, p. 442).⁶ Managers have also complained that "companies can become more reluctant to take business risks, for each time a business fails, subject to a suit for fraud."⁷

The "pressure hypothesis" is in line with the studies that emphasize the adverse effects resulting from "frivolous" shareholder lawsuits.⁸ Shareholder lawsuits are frequently instituted because self-interested attorneys urge the shareholders to file them, with only minimal evidence of breach of fiduciary duty (Romano 1991). The main beneficiaries are the plaintiff's attorneys instead of shareholders, and the firm's normal business is impeded (Swanson 1992, Rhode 2004). The cost of such suits can be enormous. Shareholder litigation distracts managers' attention, incurs settlement fees, damages a company's reputation, and results in a higher financing cost (Fich and Shivdasani 2007). Shareholder litigation, therefore, may also lead to career concerns and creates a "managerial myopia" problem (Stein 1988, 1989). Although not every firm will suffer a shareholder suit, shareholders do have the right to institute a lawsuit whenever needed. Thus, managers who are sensitive to shareholder litigation have incentives to engage in policies that reduce their legal exposure. To avert the cost incurred by litigation, managers are more likely to play it safe and encourage risk avoidance instead of far-sighted innovation (Block et al. 1993, Kinney 1994, Manso 2011).

It is empirically challenging to establish the relationship between shareholder litigation and corporate innovation. The threat of shareholder suits affects internal managerial incentives for innovation activities. However, failures resulting from innovation strategies may also trigger shareholder litigation. Our empirical investigation relies on the staggered adoption of the universal demand (UD) laws that generate intertemporal reductions in litigation risk among firms. Between 1989 and 2005, 23 states passed the laws that raise the barriers for shareholders to file derivative lawsuits against a firm's management (Davis 2008, Appel 2015). Specifically, individual shareholders can initiate a derivative lawsuit against the managers on behalf of the firm if they suspect a breach of fiduciary duty. The UD laws require that, for each derivative lawsuit, the plaintiff shareholder must first make a demand on the board of directors to take remedial action. As being alleged by the plaintiff shareholders in the claim, the board members would rarely accept such a demand, eventually leading to the dismissal of the case (Swanson 1992). Therefore,

the "universal demand requirement" has made it more difficult for shareholders filing a derivative lawsuit against the managers. Appel (2015) confirms that enforcement of the UD laws has effectively reduced the occurrence of derivative lawsuits filed by shareholders.⁹ Although the UD laws may not be randomly assigned, the staggered adoption process across states enables us to apply a difference-in-differences approach to understand the relationship between shareholder litigation and corporate innovation.

We use a propensity score matching approach to construct a sample of treated firms and control firms that shared similar traits before the laws were implemented. The treated firms are those incorporated in states that adopted the laws between 1976 and 2006. We match the treated firms with a group of control firms that are headquartered in the same state as the corresponding treated firms but are incorporated in states without the laws. By doing so, we can compare the responses of firms within the same headquarter state but subject to different legal institutions regarding shareholder litigation mandated by laws at the incorporation state level. Using a matched sample that contains 19,096 firm-year observations of U.S. public firms, we find evidence consistent with the "pressure hypothesis."¹⁰ The treated firms invest more in innovation in terms of research and development (R&D) expenditure after the adoption of the laws. The laws also lead to more explorative innovation measured by patents in unfamiliar technological classes and patents based on new knowledge. Finally, we find that, following the passage of the laws, the treated firms tend to generate more impact patents measured by forward citations and in general, achieve higher patent value. The results imply that limiting managerial discretions through shareholder litigation impedes explorative innovation activities and ultimately, has a detrimental effect on shareholders.

We conduct a battery of empirical tests to alleviate endogeneity concerns and other robustness issues. First, although the passage of a law is believed to be an endogenous process, we do not find evidence that innovative measures reversely trigger the adoption of UD laws. Second, we control for the contemporaneous changes in corporate governance by including the governance index as in Gompers et al. (2003) and institutional ownership because recent studies suggest that shareholder litigation alters corporate governance (Ferris et al. 2007, Appel 2015). Lastly, we show that our results are robust to the inclusion of additional firm characteristics, two-way clustering standard errors, and an alternative definition of the patent-based dependent variables.

We further investigate the heterogeneous impact of the laws and explore the underlying channels. We find that the effects of the reduced litigation risk on

explorative innovation are concentrated in industries with greater perceived litigation risk when proxied by the share of alleged firms in an industry. We also find that the effect is primarily driven by firms with lower institutional ownership before the ruling.

This study provides novel evidence of the influence of shareholder litigation on innovation and makes several contributions to the literature. First, this study adds to the law and finance research that investigates the importance of securities laws and shareholder protection in capital market development. Previous works typically focus on the positive effects of laws protecting the rights of shareholders (La Porta et al. 1998, 2000, 2006; Djankov et al. 2008; Brown et al. 2017). For instance, Brown et al. (2013) find that markets with stronger shareholder protection demonstrate higher R&D investment and innovation. We, however, consider shareholder litigation as a key shareholder protection mechanism rather than a more general rule of law, and we identify circumstances under which shareholder protection rights can stifle corporate innovation. By examining the adoption of laws at the state level, our study also relates to research regarding the effect of public policies on innovation and growth (e.g., Aghion et al. 2001; Bloom et al. 2002, 2016; Aghion and Howitt 2008; Lerner 2009; Acemoglu et al. 2012).

Second, our study contributes to the debate about the role of the capital market in motivating innovation. Recent empirical studies document various positive and negative determinants of corporate innovation,¹¹ including institutional ownership (Aghion et al. 2013), leveraged buyouts (Lerner et al. 2011), banking and financing factors (Brown et al. 2009, Cornaggia et al. 2015, Acharya and Xu 2017, Gu et al. 2017), nonexecutive employee stock options (Chang et al. 2015), analyst coverage (He and Tian 2013), corporate transparency (Brown and Martinsson 2019), venture capital (Chemmanur et al. 2014), credit default swaps (Chang et al. 2019), stock liquidity (Fang et al. 2014), labor unions (Bradley et al. 2017), and supply chains (Chu et al. 2019). Shareholder litigation, in contrast, mainly occurs when other governance mechanisms fail in their monitoring roles (Romano 1991). We, therefore, explore how the unique aspects of shareholder litigation as a shareholder protection mechanism impede or incentivize innovation.

Third, our evidence is related to the debate about the merit of shareholder litigation.¹² Some studies highlight the deterrence effect of shareholder lawsuits (Reinert 2014). Houston et al. (2018) find that reduced litigation risk leads to a higher cost of capital, partly because of the deterioration in financial reporting quality. Other studies, in contrast, suggest the potential

“dark side” of shareholder litigation, which might alter managerial incentives. The agency costs rooted in the shareholder litigation process may lead to numerous lawsuits with little legal merit (Fulop 2007). These lawsuits are not usually in the best interests of the shareholders because they distract managers and disrupt normal business. According to William R. McLucas, Director of the Securities and Exchange Commission (SEC) Division of Enforcement, “the SEC has acknowledged the detrimental impact of meritless securities cases. To the extent that these claims are settled to avoid litigation, they impose a tax on capital formation” (Seligman 1994, p. 439). To mitigate this concern, controlling meritless lawsuits has become a nationwide trend in recent years. Both the UD laws and the Private Securities Litigation Reform Act (PSLRA) are intended to act as a barrier to abusive lawsuits brought by shareholders (Buxbaum 1980, Swanson 1992). However, academics hold different opinions about these policies. Some believe that they have fulfilled their purpose, whereas others argue that they have unintended negative consequences, such as the deterioration of corporate governance (Johnson et al. 2007, Appel 2015). In this study, we offer the first suggestive evidence that regulations that restrict the shareholder rights to litigate against the management could encourage innovation.

This study proceeds as follows. Section 2 provides the hypothesis development, institutional details, and identification strategy. Section 3 presents the sample construction and the definitions of the key variables. The empirical results are provided in Section 4, and Section 5 concludes the paper.

2. Institutional Background and Empirical Design

2.1. Related Literature and Hypothesis Development

This section first reviews theoretical works regarding the relationship between shareholder litigation and innovation activities. Those theories can be distilled down to two conflicting hypotheses. “Disciplining hypothesis” rests on the assumption that shareholder litigation increases oversights on managers and therefore, aligns the managers’ incentives with the shareholders’ interests. In contrast, the “pressure hypothesis” argues that shareholder lawsuits might not have a positive monitoring effect but instead, that they restrict managerial discretions and exaggerate agency conflicts within firms. The differences between these two hypotheses stem from the specific institutional details in the legal mechanism of shareholder lawsuits.

2.1.1. Law and Finance. Shareholder suits are a type of legal arrangement aimed at protecting the interests of minority shareholders as discussed in the law and finance literature (e.g., La Porta et al. 1998). Managers and directors owe fiduciary duties to their shareholders, meaning that legally managers should ensure that they serve in the best interests of the shareholders. In the United States, if managers maximize their own interests at shareholders' expense, shareholders may file lawsuits for such wrongdoing. Litigation imposes personal liability on officers and directors if they are found to have breached their fiduciary duties (either duty of care or of loyalty). Prior studies document that liability rules and private enforcement help resolve the agency conflict between shareholders and the management, thereby affecting external finance and corporate governance (La Porta et al. 1997, 2000). If shareholder litigation facilitates shareholder protection, it should also promote innovation, which creates shareholder value in the long run. This prediction supports the "disciplining hypothesis."

However, legal scholars have argued that litigation may not be always instituted in the best interests of shareholders. As discussed, self-interested attorneys may urge shareholders to file meritless lawsuits with the hope of winning the attorney fees. Prior studies have documented the detrimental impact of those frivolous lawsuits on the normal business operation (Romano 1991, Swanson 1992, Rhode 2004). Regulators have implemented reforms to mitigate the concerns, such as the PSLRA. These ongoing discussions are consistent with the "pressure hypothesis" that shareholder litigation might discourage innovation. We will provide more details when we turn to the process of derivative lawsuits.

2.1.2. Agency Theory. Agency problems arise between management and shareholders because managers may not engage in activities that are in the best interests of the shareholders. Managing innovation is commonly associated with high agency costs, particularly for listed companies (Holmstrom 1989, Bernstein 2015). Both the disciplining hypothesis and the pressure hypothesis can be explained by agency theory but in different ways. The disciplining hypothesis implies that shareholder litigation can help ensure that managers and directors fulfill their fiduciary duties and shift the management away from shirking and self-dealing activities to actions that create long-term value. In contrast, the pressure hypothesis suggests that shareholder litigation may reinforce the management's capital market concerns and impede innovation. We next discuss some insights from agency theory with implications for innovation.

First, managers are averse to risky investments. Unlike other investment behavior, innovation is intrinsically risky because it involves a probability of failure and potential extraordinary outcomes. Innovation is typically a multistage process, and future contingencies are difficult to predict (Holmstrom 1989). Agency theory thus suggests that undertaking risky tasks, such as innovation, requires intensive monitoring. The disciplining hypothesis is based on the assumption that the threat of shareholder litigation exerts a monitoring effect.

Second, capital markets can induce myopic managerial behavior and stifle innovation. As discussed, innovation failure will be reflected in stock prices, leading to a high probability of shareholder litigation. The lawsuits will, in turn, send stock price tumbling and elevate subsequent financing costs. Strong track records are invaluable to firms because investors mainly rely on past performance to extrapolate firm's future for fund allocation. Managers could choose projects with early returns or choose projects with slow paybacks. As laid out in Holmstrom (1989) and Stein (1989), when both the market and managers have the same information on project option set and return distribution, whereas the managerial decision is not observed directly by investors, the management tends to choose projects that generate short-term results. The pressure hypothesis predicts that, if shareholder litigation gives rise to severe capital market concerns, innovation projects will not be undertaken sufficiently.

Such disincentive for innovation can also be formulated in the theory of career concerns, such as in Holmstrom (1999). A manager gains personal benefits by retaining her job, and potential innovation failure may put her career at risk. Diamond (1989) also highlights the importance of reputation built on past performance for firms undertaking projects financed by debt. When firms have an established reputation and enjoy low interest rates, they are more inclined to reject risky projects that involve innovation.

Third, agency theory has implications for the direction of innovation activities. Managers may weigh up the level of uncertainty in innovation projects against the benefits (Holmstrom 1989). Routine innovations can generate lower average returns but may be preferable to explorative search with high levels of uncertainty. Agency theory implies that pursuing overlapping innovations can still be beneficial to managers given the capital market concerns. Managers may be averse to challenging innovation projects, which may impose substantially private costs on managers, consistent with the "quiet life" hypothesis proposed by Bertrand and Mullainathan (2003).

Therefore, the two hypotheses suggest different strategies for firms engaging in innovation activities. The disciplining hypothesis predicts that, without the proper oversight provided by shareholder litigation, managers will rationally pass up explorative innovation projects, whereas the pressure hypothesis implies that elevated capital market concerns owing to potential litigation will impede explorative innovation and make incremental innovation preferable.

2.1.3. Past Related Empirical Studies. The disciplining hypothesis is in line with studies that highlight the effect of investor oversight on innovation. For example, Aghion et al. (2013) suggest that the career concerns about innovation can be mitigated by increased monitoring from institutional investors. Brav et al. (2018) find that the intervention of activist hedge funds enhances innovation efficiency. The pressure hypothesis is supported by studies emphasizing the adverse impact of capital market concerns on innovation. For example, Lerner et al. (2011) suggest that leveraged buyouts relieve managers of the short-term pressure from public shareholders and encourage innovation. He and Tian (2013) suggest that innovation decreases with analyst coverage because analysts can exert pressure on managers to meet short-term goals. Balsmeier et al. (2017) find that more independent boards lead to more exploitative innovation activities.

2.2. Shareholder Derivative Suits

Shareholder judicial proceedings can be divided into two main categories of direct suits and derivative suits. A direct lawsuit is brought to remedy one shareholder or a subset of shareholders (Ferris et al. 2007). For example, multiple shareholders of a defined “class” could commence a class action against a firm’s management, seeking compensation for common damages in a specific period. The other type of claims from shareholders, derivative suit, is the focus of this paper.

A shareholder derivative lawsuit is a legal action instituted by individual shareholders on behalf of the firm against their officers and directors for alleged wrongdoing that is harmful to the entire corporate entity.¹³ This type of lawsuit is derivative because the misconduct first harms the firm and then, leads to the deterioration of all shareholders’ welfare. Thus, shareholders who file derivative lawsuits are doing so on behalf of the firm instead of themselves. In the case of Tesla, the shareholder Ross Weintraub filed the lawsuit derivatively on behalf of the firm. Unlike in class actions, in derivative actions, monetary recovery is paid to the company treasury rather than the plaintiff shareholders. The importance of derivative suits is recognized in the law and finance literature. For example, La Porta et al. (1998, pp. 1114 and 1128)

state that “the rights attached to securities become critical when managers of companies act in their own interest . . . Some countries give minority shareholders legal mechanisms against perceived oppression by directors . . . These mechanisms may include the right to challenge the directors’ decisions in court (as in the American derivative suit).” In typical cases in the United States, corporate policies that trigger derivative lawsuits include value-destroying investment decisions, information-related activities, and other issues about mismanagement (Ferris et al. 2007).¹⁴ Emerging economies, such as India and China, have also instigated laws regarding shareholder derivative suits (Scarlett 2011).¹⁵

Most of the large listed firms carry liability insurance for their directors and officers to cover probable legal settlement costs. It is well documented that directors and officers (D&O) insurance protects the firm’s director and officers from personal liability in the event of litigation and could induce moral hazard problem (Lin et al. 2011, 2013). In most derivative suits, the settlement is funded or partially funded by D&O insurance. However, (D&O) insurance typically cannot cover misconduct involving dishonesty or intentional wrongdoings (Ferris et al. 2007).¹⁶ Even if a firm’s managers are not required to personally pay the settlement fees, they will still face severe punishments from the reputation damage in the labor market (Fich and Shivdasani 2007). Firms will also face heightened financing costs and stricter financing terms (Deng et al. 2014).

Derivative suits publicize the agency problems within the firm and therefore, deter directors and officers from engaging in management misconducts in the future. However, these legal actions from shareholders are also accompanied by major concerns among researchers regarding the legal merits of these claims (Fischel and Bradley 1985, Romano 1991).¹⁷ As discussed, the lawsuits are often driven by self-interested attorneys (Brandi 1993), and the detrimental impact of lawsuits without merit is well documented in the literature. As indicated by the 1995 report from Congress, the shareholder litigation system should not “be undermined by those who seek to line their own pockets by bringing abusive and meritless suits.” An abusive derivative lawsuit not only wastes a firm’s assets but also, deters the management from risk taking and experimenting through new ideas (Kinney 1994). The prevalence of excessive litigation may induce officers and directors to focus more on legally safe activities rather than on far-sighted innovation, thereby harming the competitiveness of the whole economy (Block et al. 1993).

2.3. Universal Demand Laws

A derivative suit in the United States proceeds through several steps. Before bringing a derivative

action, the plaintiff shareholders must first demand that their board take action to address the alleged concerns. This process is called the “demand requirement.” The board can choose to reject, consider, or ignore the request within a reasonable time. However, in reality, directors almost always reject the demand because the board members are usually those targeted. Shareholders can still proceed with the derivative suit after the demand is refused or unanswered. However, if the demand is rejected, in most cases the court follows the board’s decision and dismisses the claim pursuant to the business judgment rule.

Shareholders, however, can circumvent the demand requirement by arguing the futility of demand if they can provide evidence showing that the board of directors cannot fairly evaluate it.¹⁸ In practice, shareholders prefer to plead the futility exception because it is difficult to proceed with a lawsuit if the board refuses the demand. In the case of Tesla, the shareholder argued that making the demand would be futile because “current members of Tesla’s Board are antagonistic to this lawsuit.”

Between 1989 and 2005, 23 states in the United States implemented the UD laws, which impose the demand requirement on every derivative lawsuit filed in these states. After the enactment of the laws, shareholders are deprived of the option to plead demand futility. As illustrated in the online appendix, Table IA 1, the first states to adopt the laws were Georgia and Michigan in 1989, and the most recent were Rhode Island and South Dakota in 2005. The idea behind the UD laws comes from the Model Business Corporation Act, a uniform law proposed by the American Bar Association that is voluntarily followed by some states. Because the UD laws require plaintiffs to make a demand as a prerequisite to filing a derivative suit and because the demand would be refused in most cases, the universal demand requirement serves as a significant barrier to filing derivative lawsuits. Consistent with the findings of Appel (2015), we find that the number of shareholder derivative lawsuits has significantly dropped since the UD laws were first adopted.

2.4. Identification Strategy

Firms incorporated in states that have passed the UD laws are relatively insulated from shareholder derivative lawsuits for the reasons discussed. We exploit these incorporation state-level shocks as natural experiments to understand the relation between shareholder litigation and innovation. This setting has several features that facilitate a difference-in-differences analysis. First, the staggered adoption of the UD laws generates intertemporal and cross-state variations in

litigation risk *ex ante*. Second, similar to Bertrand and Mullainathan (2003) in the case of the Business Combination Laws, the legal shocks are at the incorporation state level. Empirically, this feature allows us to compare firms that are headquartered in the same state but are subject to different legislations. This empirical design thus mitigates the confounding effects resulting from regional economic shocks.

Our difference-in-differences specification is as follows:

$$Innovation_{it} = \alpha + \beta Law_{it} + \gamma X_{it} + \theta_i + \delta_{it} + \varepsilon_{it}, \quad (1)$$

where $Innovation_{it}$ is the innovation measure gauged by several proxies. Law_{it} equals one if the incorporation state of the firm has a UD law. β is the main coefficient of interest for identifying the effect of the laws. θ_i denotes the firm fixed effects that capture the firm-level time-invariant effects. δ_{it} represents the headquarter state by year fixed effects that pick up regional trends. X_{it} denotes lagged firm-level attributes, including firm size, leverage, book-to-market ratio, firm age, and profitability. We also include lagged R&D expenditure as a control variable except in the regression using R&D as the dependent variable.

Like the enactment of other laws, the staggered adoption of UD laws may not be random. Economic, political, or other unobservable factors can contribute to the spread of the laws. The legislative process usually involves public and legislature-level debate and private lobbying by both interest groups and large public firms. One may particularly worry about the political economy of the laws. For example, the adoption of takeover laws is influenced by large business groups in the economy (see Romano 1987). It is possible that firms troubled by frivolous derivative lawsuits engage less in the innovation process and lobby for the adoption of UD laws. Although we cannot eliminate the associated concerns, in the following we conduct several empirical tests and discuss to what extent the endogeneity issues might affect the interpretations of our findings.

3. Sample and Variables

3.1. Sample Selection

The data set in our study consists of data from several sources. We first collect information on firm characteristics from Compustat. Like other corporate laws (e.g., Bertrand and Mullainathan 2003), the effect of the UD law is at the incorporation state level, and therefore, firms incorporated in states with effective UD laws are treated. Firms can, however, change their state of incorporation in the process of doing business. For valid inference, it is important to correctly

identify a firm's historical state of incorporation. Compustat only provides data on the latest state of incorporation, which can create measurement errors in variable construction. Thus, we use the historical state of incorporation provided by Bill McDonald, who compiled each firm's state of incorporation based on its original SEC filing since 1994.¹⁹ We supplement this information with Compustat records in the case of missing values.

We consider a list of basic firm attributes. We include firm size (*Size*) and the market-to-book ratio (*MTB*) because the size of a firm and its growth opportunities are likely to correlate with innovative activities. We use leverage (*Leverage*) to account for the extent of financial constraints because financial distress may affect a firm's propensity to innovate. In addition, we control for log value of firm age ($\ln(\text{age})$) because older firms may search in older technological areas. The firm's profitability (measured by return on assets (*ROA*)) is also considered. In the robustness check, we include the governance index (*G index*) and institutional ownership (*IO*) as proxies for corporate governance. Other variables on firm characteristics will be discussed when used.

Like Bertrand and Mullainathan (2003), we exploit the differences in legal arrangements among firms headquartered in the same state. To assure that the treated firms and control firms are similar along basic firm characteristics, we construct a matched sample based on a propensity score matching approach. Specifically, we first identify 471 treated firms incorporated in states that adopted the UD laws during our sample period. We then match each treated firm with a control firm through propensity score matching based on the covariates of R&D expenditure, firm size, leverage, market-to-book ratio, firm age, and profitability in the year before the law effective year. We restrict the control group to include firms incorporated in states that never adopted the law during our sample period but headquartered in the same state as the corresponding treated firms. Both the treated firms and the control firms have financials before and after the law enactment. The final sample includes 942 unique firms and 19,096 firm-year observations from 1976 to 2006.

3.2. Innovation Variables

Our innovation measures are categorized as innovation inputs, types of innovation activities, and the economic value of innovation. First, we use R&D expenditure from Compustat to measure a firm's investment in innovation. The variable *R&D/Assets* is the value of R&D expenses scaled by total assets. Missing R&D values are treated as zero. Second, we

use patent information to assess different innovation activities. The patent information is extracted from the National Bureau of Economic Research (NBER) database, which covers the period from 1976 to 2006 and provides links matching the patent assignees to firm identifiers in Compustat. *Patent* is the total number of patents, which can be of high or low quality. We follow Balsmeier et al. (2017) and construct variables to differentiate explorative innovation and exploitive innovation. We first construct two variables based on the relationship between a firm's innovation strategy and its experience in specific technological patent classes. *New class Patent* is the number of patents filed in a given year in technology classes where the firm has no previous patents filed. *Known class Patent* is the number of patents filed in a given year in a technology class previously known to the firm. Firm's efforts to produce more patents in new technological areas indicate the presence of more explorative innovation activities. In contrast, firms that file more patents in familiar technological classes may be more likely to apply their previous patent experience and avoid explorative innovation search.

We further construct variables taking a firm's current patent knowledge based on backward citations into consideration. A patent is considered as explorative if at least a defined percentage of the citations that it refers is *not* from the firm's existing knowledge. Here existing knowledge includes all of the patents that the firm produced in the history and patents that were cited by the firm's patents filed over the past five years (Brav et al. 2018). We define *New knowledge Patent* as the number of patents filed with at least 90% of the backward citations not from existing knowledge. Similarly, we define *Known knowledge Patent* as the number of patents in which at least 90% of their knowledge that they refer to is from existing knowledge. These firm-level aggregated variables indicate whether a firm focuses on explorative search or exploits knowledge that it already has.

Next, we categorize the patents according to their position in the distribution of forward citations in a given technological class in an application year. *Top10% Patent* is a firm's total number of patents in the top 10% of the most cited among patents filed by listed firms in a given technological class and application year. We also define *Top5% Patent* as the number of patents in the top 5% of the citation distribution. Both measures quantify the high-quality innovations of a firm.

Last, we infer the economic value of a patent from the market's reactions to the announcement of the granting of the patent following Kogan et al. (2017). We estimate three measures of the patent value variables based on different assumptions. The variables

are defined as the total value of patents applied by a firm scaled by market capitalization in the application year, and the detailed construction is in Section 4.4.

We follow the corporate innovation literature (e.g., Hsu et al. 2014) and address two types of well-documented truncation problems associated with the NBER patent database. The first is because of the application-grant lag in the patent granting process. We only observe patents granted up until 2006, but it takes two years on average for a filed patent to be eventually granted. Thus, many patent applications may still be under review, and we observe a decrease in the number of granted patents in the last years of our sample period (2005 and 2006). We follow Hall et al. (2001, 2005) to address this truncation problem when counting the number of granted patents. We obtain a series of weight factors using the empirical distribution of the application-grant gap, and we adjust our measures based on patent counts with the constructed weights. Second, truncation problems regarding patent forward citations also occur in the NBER database. Patents continue to receive citations over long periods, and the NBER database only allows us to observe citations up to 2006. We address this type of truncation by updating citation data using PatentsView, which contains patent citations until May 2019.

Following the practices in the literature, we use the natural logarithm of these patent-based variables in the regression analysis to mitigate concerns about skewness and to facilitate a reasonable econometric interpretation. We also add one to the actual number in calculating the logarithm value to include the firm-year observations with zero patents in our analysis. The correlation matrix for innovation measures is provided in the online appendix, Table IA 2.

Tables 1 and 2 present the descriptive statistics of the main variables. Table 1 shows the summary statistics of our main sample matched on R&D expenditure, firm size, leverage, market-to-book ratio, firm age, and ROA. Continuous variables are winsorized at 0.5% to correct for the effect of potential outliers. The balance tests presented in Table 2 indicate that the matching variables do not demonstrate significant differences between the treated group and the control group.

4. Results

In this section, we first present evidence for the validity of our identification strategy, and then, we examine the results concerning the relationship between shareholder litigation and corporate innovation along several dimensions, specifically focusing on explorative innovation activities. The robustness checks and the tests on the heterogeneous impact of shareholder

Table 1. Summary Statistics: Descriptive Statistics

Variable	N	Mean	SD
<i>Law</i>	19,096	0.211	0.408
<i>R&D/Assets</i>	19,096	0.030	0.087
<i>ln(Patent)</i>	19,096	0.430	1.002
<i>ln(New class Patent)</i>	19,096	0.186	0.470
<i>ln(Known class Patent)</i>	19,096	0.352	0.970
<i>ln(New knowledge Patent)</i>	19,096	0.314	0.789
<i>ln(Known knowledge Patent)</i>	19,096	0.153	0.558
<i>ln(Top 10% Patent)</i>	19,096	0.117	0.443
<i>ln(Top 5% Patent)</i>	19,096	0.068	0.318
<i>Patent Value1</i>	19,096	0.023	0.070
<i>Patent Value2</i>	19,096	0.022	0.069
<i>Patent Value3</i>	19,096	0.022	0.067
<i>Size</i>	19,096	4.937	2.267
<i>Leverage</i>	19,096	0.503	0.407
<i>MTB</i>	19,096	1.755	2.167
<i>ln(age)</i>	19,096	2.600	0.676
<i>ROA</i>	19,096	0.095	0.248

Notes. This table presents the summary statistics of the main variables used in this study. The sample includes 942 unique firms and 19,096 firm-year observations from 1976 to 2006. We first identify 471 treated firms incorporated in states that eventually adopted universal demand laws during our sample period. We then match each treated firm with a control firm using a propensity score matching approach based on a list of covariates, including R&D expenditure, firm size, leverage, market-to-book ratio, firm age, and profitability in the year prior to the first law effective year. Throughout our sample, we restrict the control firms to those incorporated in states without the law but headquartered in the same state as the corresponding treated firms. This table reports the descriptive statistics. All variables are defined in the appendix. Continuous variables are winsorized at the 0.5% level. SD, standard deviation.

litigation are also provided. Finally, we discuss the possible interpretations for our results.

4.1. Setting Validity

The difference-in-differences design rests on the parallel trend assumption, and the shocks at hand are exogenous. However, we cannot directly eliminate exogeneity concerns regarding the implementation of the law, and therefore, we check for obvious evidence of reverse causality in our design.

Table 3 suggests that we do not find that the timing of a state's enactment of UD laws is mainly driven by previous innovation measures. We apply a Weibull hazard model (Beck et al. 2010) in which the dependent variable is the log of the time expected until the passing of the laws. The sample is from 1976 to 2006 and at the incorporation state-year level. The explanatory variable corresponds to the key measures of innovation aggregated at the state level. For example, $R\&D/Assets_t$ is the average R&D to assets ratio for firms incorporated in a state in year t , whereas $R\&D/Assets_{t-1}$ is the variable measured in year $t - 1$. $\Delta R\&D/Assets_t$ is the change in the average R&D to assets ratio for firms incorporated

Table 2. Summary Statistics: Balance Tests

	(1)	(2)	(3)	(4)
Variable	Treat mean	Control mean	Difference	p-Value
<i>R&D/Assets</i>	0.028	0.035	−0.007	0.195
<i>Size</i>	4.887	4.767	0.121	0.417
<i>Leverage</i>	0.500	0.526	−0.026	0.232
<i>MTB</i>	1.749	1.896	−0.147	0.271
<i>ln(age)</i>	2.703	2.710	−0.007	0.850
<i>ROA</i>	0.100	0.082	0.018	0.102
<i>N</i>	471	471		

Notes. This table presents the summary statistics of the main variables used in this study. The sample includes 942 unique firms and 19,096 firm-year observations from 1976 to 2006. We first identify 471 treated firms incorporated in states that eventually adopted universal demand laws during our sample period. We then match each treated firm with a control firm using a propensity score matching approach based on a list of covariates, including R&D expenditure, firm size, leverage, market-to-book ratio, firm age, and profitability in the year prior to the first law effective year. Throughout our sample, we restrict the control firms to those incorporated in states without the law but headquartered in the same state as the corresponding treated firms. The balance tests of the matching variables are presented. All variables are defined in the appendix. Continuous variables are winsorized at the 0.5% level.

in a state from year $t - 1$ to year t . Other measures are defined similarly. We control for several state-level characteristics to pick up the contemporaneous effects related to the regional economy, such as state-level real gross domestic product (GDP), GDP growth, and the number of firms incorporated in a state. As shown in Table 3, the coefficients of the state-level innovation measures are insignificant. Thus, we cannot reject the null hypothesis that firm-level corporate innovation does not affect the timing of adopting UD laws. It is worth noting that, for firms engaging in lobbying activities, they could hold investments in innovation projects until the passage of the laws. In Section 4.8, we will further discuss how the endogeneity concerns affect our interpretations.

We next verify that the adoption of UD laws indeed leads to fewer shareholder derivative lawsuits in practice. We collect legal cases from Audit Analytics and identify shareholder derivative lawsuits as those classified as “shareholder suits” and “derivative.” The resulting sample contains about 500 derivative cases between 2000 and 2013 because Audit Analytics only contains lawsuits filed after 2000. During the sample period, there are five states that adopted UD laws. We run the difference-in-differences regression at the incorporation state level to demonstrate the impact of UD laws on shareholder derivative actions. As Model (1) of Table 4 shows, the number of derivative suits drops by 14% following the adoption of UD laws. When controlling for economic

factors, the coefficient is still significantly negative. The findings are consistent with Appel (2015), suggesting that UD laws indeed raise the barrier for derivative litigation and reduce the associated litigation threat for managers.

Another concern is that, by increasing the procedural hurdles of a derivative lawsuit, UD laws may, in theory, encourage shareholders to bring direct actions. If so, the number of securities class actions by firms incorporated in a state would increase following the law adoption. To test this hypothesis, we collect the data on class actions from the Stanford Securities Class Action Clearinghouse. As the data are first collected in 1996, we consider the period from 1996 to 2013. The number of class actions is aggregated at the incorporation state level. The results reported in Models (3) and (4) of Table 4 suggest that the number of class actions does not significantly change after the adoption of UD laws. The coefficient remains indifferent from zero after controlling for state real GDP and GDP growth rate. Thus, we cannot reject the null hypothesis that UD laws do not lead to more class actions, suggesting that UD laws do reduce managers’ overall exposure to shareholder litigation. Several explanations for this pattern can be considered. First, as the Federal Rule of Civil Procedure states, there are several prerequisites for filing shareholder class actions. For example, Rule 23 (a) requires that “the class is so numerous that joinder of all members is impracticable.”²⁰ These requirements potentially prevent the transition from derivative actions to class actions. Second, class and derivative actions may have different underlying motivations. In derivative actions, only the attorney’s fee can be recovered through winning the lawsuit, whereas any monetary recovery will flow to the firm instead of the plaintiff shareholders. Because of this reason, prior studies suggest that derivative litigation is partially driven by the goal of recovering the attorneys’ fees instead of legal merit. If so, the passing of UD laws undermines this motivation by raising the procedural hurdle, but it will not necessarily give rise to more class actions in which plaintiff shareholders can be directly compensated.

“Incorporation state shopping” may also be a concern. Because UD laws increase the barriers for shareholder derivative litigation against management, firm managers may aim to change their incorporation state to one with UD laws, thus mitigating concerns over the threat of litigation. We conduct an empirical analysis to assess this possibility. The results in Models (5) and (6) of Table 4 suggest that the passage of UD laws does not significantly alter the number of firms incorporated in a state.²¹

Table 3. Validity Tests for the Natural Experiment: Timing of Universal Demand Litigation and Preexisting Innovation Activities

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$R\&D/Assets_t$	3.371 (2.575)								
$\ln(Patent)_t$		0.108 (0.321)							
$Patent\ Value1_t$			6.169 (6.290)						
$R\&D/Assets_{t-1}$				3.947 (2.548)					
$\ln(Patent)_{t-1}$					0.165 (0.356)				
$Patent\ Value1_{t-1}$						5.288 (6.264)			
$\Delta R\&D/Assets_t$							-0.507 (0.700)		
$\Delta \ln(Patent)_t$								-0.171 (0.495)	
$\Delta Patent\ Value1_t$									0.197 (2.404)
$\ln(GDP)$	0.062 (0.075)	0.051 (0.080)	0.044 (0.081)	0.060 (0.075)	0.048 (0.081)	0.044 (0.080)	0.053 (0.078)	0.052 (0.078)	0.054 (0.078)
$GDP\ Growth$	-0.025 (0.017)	-0.025 (0.017)	-0.025 (0.017)	-0.024 (0.017)	-0.025 (0.017)	-0.025 (0.017)	-0.026 (0.017)	-0.026 (0.017)	-0.026 (0.017)
$\ln(\# \text{ of incorporated firms})$	-0.029 (0.048)	0.004 (0.045)	-0.010 (0.048)	-0.032 (0.046)	0.001 (0.046)	-0.007 (0.047)	0.009 (0.043)	0.008 (0.042)	0.008 (0.043)
Observations	1,238	1,238	1,238	1,238	1,238	1,238	1,238	1,238	1,238

Notes. This table presents the results of the validity tests for the natural experiment in this study. The model is a Weibull hazard model where the dependent variable is the log expected time for the enforcement of UD laws. The sample is at the incorporation state-year level and from 1976 to 2006. The independent variables are measures of firms' innovation activities aggregated at the incorporation state and year level. For example, $R\&D/Assets_t$ is the average R&D to assets ratio for firms incorporated in a state in a given year t . $R\&D/Assets_{t-1}$ is the average R&D to assets ratio for firms incorporated in a state in year $t - 1$. $\Delta R\&D/Assets_t$ is the change in the average R&D to assets ratio for firms incorporated in a state from year $t - 1$ to year t . GDP and GDP growth rate for each incorporation state are from the Bureau of Economic Analysis, U.S. Department of Commerce. The number of incorporated firms is the number of public firms incorporated in each state. The standard errors are clustered at the incorporation state level, and they are given in parentheses.

4.2. Innovation Input

We investigate firms' responses in terms of R&D investments following the adoption of UD laws. The disciplining hypothesis predicts a negative coefficient, whereas the pressure hypothesis predicts a positive significant coefficient. However, the pressure hypothesis cannot be rejected even if the coefficient is not significant. Managers may steer their innovation strategy toward riskier projects while keeping the innovation input unchanged.

Following the literature, we use R&D expenditure scaled by total assets as a measure of investment in innovation. Table 5 presents the empirical results. Model (1) of Table 5 is a standard ordinary least squares model with the firm and headquarter state by year fixed effects without any endogenous control variables. This model reveals a standalone effect of the UD laws on R&D investment, which avoids the "bad control" problem (Angrist and Pischke 2008). The treatment effect is quantified by the coefficient

associated with *Law*. Consistent with Bertrand and Mullainathan (2003), the standard error is clustered at the incorporation state level. As Model (1) of Table 5 shows, the treated firms on average invest about 0.3% more in R&D than firms incorporated in states without UD laws. The increase accounts for about 10% of the sample mean for R&D expenditure, indicating an economically meaningful effect. We have controlled for the effect of regional confounding factors. Because we include state by year fixed effects in the regressions, the increase is measured relative to firms headquartered in the same state.

In Model (2) of Table 5, we include the firm attributes of firm size, market-to-book ratio, leverage ratio, firm age, and ROA to control for the intertemporal changes in firm fundamentals. The empirical results are robust when these attributes are included. The coefficient remains at 0.2% after the full set of controls is included, and it is still significant at the 5% level. Consistent with the pressure hypothesis,

Table 4. Validity Tests for the Natural Experiment: Universal Demand Litigation, Derivative Suits, Class Actions, and Incorporation State Shopping

	(1)	(2)	(3)	(4)	(5)	(6)
	Period: 2000–2013		Period: 1996–2013		Period: 1994–2013	
	ln(<i>Derivative Actions</i>)		ln(<i>Class Actions</i>)		ln(# of incorporated firms)	
<i>Law</i>	−0.142** (0.06)	−0.132** (0.06)	−0.010 (0.11)	−0.010 (0.11)	0.292 (0.29)	0.265 (0.29)
ln(<i>GDP</i>)		−0.680 (0.42)		0.013 (0.21)		1.412 (1.68)
<i>GDP Growth</i>		−0.007 (0.01)		−0.001 (0.00)		−0.000 (0.02)
Incorporation state FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	900	900	1,000	1,000
Adjusted R^2	0.675	0.678	0.775	0.774	0.829	0.830

Notes. This table presents the results of the validity tests for the natural experiment in this study. All variables are at the incorporation state-year level. Models (1) and (2) test whether the adoption of UD law leads to fewer shareholder derivative lawsuits. *Law* is an indicator equal to one for firms incorporated in a state in the years after the UD law is adopted. ln(*Derivative Actions*) is the log value of one plus the number of derivative lawsuits. The information on derivative lawsuits is from Audit Analytics and covers the period between 2000 and 2013. We include incorporation state and year fixed effects and control for the log value of GDP and GDP growth. Models (3) and (4) test whether the adoption of UD laws substitutes for the shareholder's class action litigation activities. The dependent variable, ln(*Class Actions*), is the log value of one plus the number of class actions. The sample of shareholder class actions is collected from Stanford Securities Class Action Clearinghouse and starts from 1996. Models (5) and (6) test the possibility of incorporation shopping following the adoption of UD laws. The dependent variable, ln(# of incorporated firms), is the log number of firms incorporated in a state. FE denotes fixed effects. The standard errors are clustered at the incorporation state level, and they are given in parentheses.

**5% significance level.

the evidence suggests that, when treated firms are less likely to be sued by shareholders through derivative lawsuits, they become more incentivized to invest in innovation.

Table 5. Effect of Universal Demand Laws on R&D Expenditure

	(1)	(2)
	<i>R&D/Assets</i>	
<i>Law</i>	0.003** (0.001)	0.002** (0.001)
Controls	No	Yes
Firm FE	Yes	Yes
Headquarter state-year FE	Yes	Yes
Observations	19,096	19,096
Adjusted R^2	0.641	0.667

Notes. This table presents the effect of UD laws on a firm's R&D expenditure. *R&D/Assets* is R&D expenditure scaled by total assets. R&D expenditure is set to zero when missing. *Law* is an indicator equal to one for firms incorporated in a state in the years after the law is adopted. Lagged firm size, leverage ratio, market-to-book ratio, firm age, and ROA are included as control variables. The definitions are given in the appendix. Firm and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level and shown in parentheses.

**5% significance level.

4.3. Patent Categories

We have established the relationship between the threat of shareholder litigation and investment in innovation, and our next concern is how the enactment of UD laws influences a firm's specific strategy in innovation. We base our analysis on several patent-based variables.

We have no strong theoretical predictions about the effect of the laws on the number of patents. The pressure hypothesis suggests that shareholder litigation limits managerial discretion and that firms produce fewer patents when the threat of litigation is present. Thus, because the UD laws reduce the litigation risk, the number of patents produced by the treated firms may increase. However, the excessive pressure of shareholder litigation may also induce managers to conduct exploitive innovation and generate low-quality patents. The adoption of UD laws will then lead to fewer patents. Therefore, both positive and negative effects are consistent with the pressure hypothesis, and which effect is dominant remains an empirical question. The results reported in Model (1) of Table 6 indicate that the UD laws do not have a significant effect on the total number of patents. In this regression, we control for lagged R&D

expenditure scaled by assets as the control variable. The evidence is consistent with our reasoning and suggests that, although the treated firms experience more research input in terms of increased R&D expenditure, no material increase in the total number of their patents occurs.

As discussed, the threat of shareholder litigation can lead to managerial myopia, which may discourage radical innovation. We formally test this possibility by considering additional measures based on patent citations. First, we assess the number of patents in technological classes that are previously unknown to the firm, which are new classes that the firm has not filed any patents in since 1976. Filing in a new class of patent indicates that the management is entering a new technological realm and that it is bearing the potential risk of failure. We define the complement as the number of patents with known classes. An increased number of known class patents indicates a high probability of pursuing exploitive innovation. Models (2) and (3) of Table 6 present the corresponding regression results suggesting that the law enactment has an insignificant effect on searching into previously patented classes but has a strong and significantly positive effect on the exploration of new classes. The number of patents in new classes increases by 1.9% following the adoption of the laws. The evidence indicates that a reduction in shareholder litigation results in a firm moving toward more explorative innovation.

Second, we examine whether UD laws lead to more explorative innovation gauged by the backward

citations of the patent portfolio. We define a patent as a “new knowledge” patent if 90% of its backward citations involve new knowledge rather than existing knowledge already held by the firm. The firm’s existing knowledge includes all of the patents filed by the firm and all patents that are cited by patents filed by the firm over past five years. The number of “new knowledge” patents filed by a firm directly indicates whether the firm’s innovation strategy is heavily dependent on past knowledge. As Model (4) of Table 6 shows, the treated firms experience a 3.2% increase in the number of “new knowledge” patents following the adoption of UD laws. We also find that the laws have no significant effect on the number of “known knowledge” patents, which are defined as patents in which 90% of the backward citations are from firm’s existing knowledge.

In summary, the evidence from several patent-based measures supports the argument that the reduced threat of litigation promotes innovation, particularly in new and explorative areas, leading to a shifted composition of patent portfolios.

4.4. High-impact Patent and Patent Value

Our results are consistent with the argument that restricted shareholder rights in filing lawsuits, possibly frivolous lawsuits, promote innovation. However, the findings do not necessarily suggest a “bright side” of the UD laws for shareholders. The law adoption may lead to excessive innovation, such as investing in risky projects, when the barrier of instituting the suit is heightened. This alternative story supports the

Table 6. UD Laws and Patent Outputs

	(1)	(2)	(3)	(4)	(5)
	ln(Patent)	ln(New class Patent)	ln(Known class Patent)	ln(New knowledge Patent)	ln(Known knowledge Patent)
<i>Law</i>	0.009 (0.024)	0.019** (0.008)	−0.005 (0.022)	0.032*** (0.012)	−0.002 (0.011)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes
Observations	19,096	19,096	19,096	19,096	19,096
Adjusted R ²	0.797	0.462	0.804	0.762	0.705

Notes. This table presents the effect of UD laws on the number of different types of patents filed in a given year. *Law* is an indicator equal to one for firms incorporated in a state in the years after the UD law is adopted. *ln(Patent)* is the natural logarithm of one plus the number of eventually granted patents filed by the firm in a given year. *ln(New class Patent)* is the natural logarithm of one plus the number of patents filed in technology classes previously unknown to the firm in a given year. *ln(Known class Patent)* is the natural logarithm of one plus the number of patents filed in a technology class previously known to the firm in a given year. *ln(New knowledge Patent)* is the natural logarithm of one plus the number of “new knowledge” patents filed in a given year. A patent is defined as a “new knowledge” patent if at least 90% of the citations that it refers do not come from existing knowledge, which includes all of the patents that the firm produced and all of the patents cited by the firm’s patents filed over the previous five years. *ln(Known knowledge Patent)* is the natural logarithm of one plus the number of “known knowledge” patents filed in a given year. A patent is defined as a “known knowledge” patent if at least 90% of the citations that it refers to come from existing knowledge. Lagged R&D expenditure scaled by assets, firm size, leverage ratio, market-to-book ratio, firm age, and ROA are included as control variables. The definitions are given in the appendix. Firm and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level and shown in parentheses.

5% significance level; *1% significance level.

disciplining hypothesis and could also explain our findings so far. To further differentiate the two hypotheses, we need to explore the shareholder implications of the laws more directly.

We first examine whether the passage of UD laws leads to more high-impact patents. We quantify high-impact patents (such as breakthrough patents) by using two variables according to the forward citations that the patents receive. As shown in Models (1) and (2) of Table 7, the UD laws lead to a significant increase in a range from 2% to 3% in patents in the top 10% of the citation distribution in a given technological class in an application year. Models (3) and (4) of Table 7 suggest that the law adoption also leads to more patents in the top 5% of this citation distribution. The results are consistent with the pressure hypothesis that lower litigation risk incentivizes innovation activities in the interest of shareholders.

We next explore the implications for shareholder wealth by using patent value, which is inferred from the market reactions to the announcement of patent granting, following Kogan et al. (2017). The market upgrades a firm when a high-quality patent is granted to it. This market-based measure has several advantages in our context. As Kogan et al. (2017) note, the stock prices are forward looking, which allows us to quantify the private value to the patent holder. Citations instead measure only the scientific value of the patent. In addition, the patent value in terms of dollars facilitates

comparison across industry and time so that we could aggregate the patent value at the firm level.

We follow Kogan et al. (2017) and define the patent value for patent j as

$$\xi_j^1 = (1 - \bar{\pi})^{-1} \frac{1}{N_j} E[v_j | R_j] M_j, \quad (2)$$

where $\bar{\pi}$ is the probability of a successful patent application, N_j is the number of multiple patents granted on the same date as patent j , and M_j is the market capitalization on the date before the announcement of the patent grant. $E[v_j | R_j]$ is the expectation of the patent value v_j conditional on the idiosyncratic return R_j , which is the three-day cumulative market-adjusted return from day 0 to day 2 around the issuance date. $E[v_j | R_j]$ is a positive value and estimated following Kogan et al. (2017), equation 4. *Patent Value1* is defined as the total value of granted patents applied in a year scaled by market capitalization. Formally,

$$\text{Patent Value1}_{i,t} = \frac{\sum_{k=1}^K \hat{\xi}_k^1}{M_{i,t}}, \quad (3)$$

where K is the total number of patents applied in year t , $M_{i,t}$ is the market value of firm i in year t , and $\hat{\xi}_k^1$ is the value of the patent in dollars for the application year adjusted using the consumer price index. In our sample, 13 patents are granted per announcement date on average. In the following, we consider the forward citations of each patent because patents granted on the same date may not be equally valuable:

$$\xi_j^2 = (1 - \bar{\pi})^{-1} w_j E[v_j | R_j] M_j, \quad (4)$$

where w_j is $c_j / \sum_{q=1}^Q c_q$. c_j is the number of citations for patent j , and c_q denotes the number of citations for patent q granted on the same date. The firm-year-level measure is defined as $\text{Patent Value2}_{i,t} = (\sum_{k=1}^K \hat{\xi}_k^2) / M_{i,t}$. We also estimate another patent value ξ_j^3 using the alternative estimation of the idiosyncratic return. Formally, $\xi_j^3 = (1 - \bar{\pi})^{-1} w_j E[v_j | R'_j] M_j$, where R'_j is the three-day cumulative abnormal return based on the market model. *Patent Value3* _{i,t} is defined accordingly.

The value of a patent is highly skewed. Breakthrough patents generated by the firm's explorative innovation are likely to be associated with a high market value. Table 8 shows that the passage of UD laws leads to a 0.4% to 0.5% increase in patent value, which accounts for roughly 6% to 7% of the standard deviation. These results suggest that the enactment of the laws significantly changes the composition of a firm's patent portfolio, leading to a higher patent value for treated firms. Thus, we find no clear

Table 7. UD Laws and Impact Patents

	(1)	(2)	(3)	(4)
	ln(Top 10% Patent)	ln(Top 5% Patent)		
<i>Law</i>	0.030*** (0.009)	0.020** (0.007)	0.016*** (0.004)	0.009*** (0.004)
Controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes
Observations	19,096	19,096	19,096	19,096
Adjusted R ²	0.677	0.680	0.610	0.612

Notes. This table presents the effect of UD laws on the production of impact patents. *ln(Top10% Patent)* is the natural logarithm of one plus a firm's total number of patents that are in the top 10% category of the distribution of forward citations for patents granted to the U.S. public firms in a given technological class in an application year. *ln(Top5% Patent)* is the natural logarithm of one plus a firm's total number of patents that are in the top 5% category of the distribution of forward citations in a given technological class in an application year. *Law* is an indicator that is equal to one for firms incorporated in a state in the years after the UD law is adopted. Lagged R&D expenditure scaled by assets, firm size, leverage ratio, market-to-book ratio, firm age, and ROA are included as control variables. The definitions are given in the appendix. Firm and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level and shown in parentheses.

5% significance level; *1% significance level.

Table 8. UD Laws and Patent Values

	(1)	(2)	(3)	(4)	(5)	(6)
	Patent Value1		Patent Value2		Patent Value3	
<i>Law</i>	0.005*** (0.002)	0.004** (0.002)	0.005*** (0.002)	0.004** (0.002)	0.005*** (0.002)	0.004** (0.002)
Controls	No	Yes	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,096	19,096	19,096	19,096	19,096	19,096
Adjusted R^2	0.626	0.628	0.622	0.624	0.624	0.626

Notes. This table presents the effect of UD laws on the economic value of patents. *Patent Value1* is the total value of patents applied in a year inferred from the market reactions on the grant date scaled by the market value of equity in the application year. *Patent Value1* is based on three-day cumulative abnormal returns from date 0 to date +2 around the grant date using the market-adjusted method following Kogan et al. (2017). The value of a patent is converted to the application year using consumer price index before being aggregated at the application year level. We then follow the logic of Kogan et al. (2017) and calculate two alternative measures, *Patent Value2* and *Patent Value3*. To define *Patent Value2*, when there are multiple patents announced on the same date, we assign the estimated economic value across patents according to their number of forward citations rather than following the equal weighting method used by Kogan et al. (2017). To define *Patent Value3*, we first estimate a market model to construct the abnormal returns around the patent announcement date. The forward citations are also used as weights when multiple patents are announced on the same date. *Law* is an indicator equal to one for firms incorporated in a state in the years after the UD law is adopted. Lagged R&D expenditure scaled by assets, firm size, leverage ratio, market-to-book ratio, firm age, and ROA are included as control variables. The definitions are given in the appendix. Firm and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level and shown in parentheses.

5% significance level; *1% significance level.

evidence that a reduced threat of litigation leads to “excessive innovation.”

4.5. Pretrends

The parallel trend assumption is central to the difference-in-differences design. The assumption indicates that, if the considered exogenous shock did not happen, the responses of treated and control firms would follow a parallel trend. Because the counterfactual situation is unobservable, the underlying assumption cannot be directly verified. As a compromise solution, we can only check the obvious evidence of pretrends across outcome variables in the period before the law adoption.

We consider a model specification with treatment dummies from three years before the UD laws become enforced until three years after. The following regression is estimated:

$$Innovation_{it} = \alpha + \sum_{k=-3}^{k=3} \beta^k Law^k + \gamma X_{it} + \theta_i + \delta_{it} + \varepsilon_{it}, \quad (5)$$

where Law^k is an indicator equal to one for the k th year relative to the UD law enforcement year, Law^{+3} is set to one for the years after the third effective year, and $Innovation_{it}$ denotes the main variables in our study. Other specifications follow the convention as in Equation (1). In general, we find no evidence to suggest that a firm’s key innovation measures are

significantly different between treated and control prior to the enactment of the law as shown in Table 9. Dynamic effects for patent values are provided in the online appendix, Figure IA 1.

4.6. Robustness Checks

Several robustness checks are presented in Tables 10–13. First, we attempt to isolate the effect of litigation risk and explicitly control for the contemporaneous effects of corporate governance. Appel (2015) finds that the UD laws lead to the deterioration of corporate governance through governance provisions and institutional ownership proxies. We thus consider the governance index (G index) introduced by Gompers et al. (2003) and institutional ownership as control variables. The data on the G index is collected from ISS (formally RiskMetrics), which dates back to 1990. We complete a firm’s G index with the nearest available data point back to 1981 to include the variation in shareholder litigation generated from the adoption of the UD laws in the 1980s. The data on institutional ownership are derived from the Thompson Reuters Institutional Holding (13F) Database. As shown in Table 10, after controlling for these two governance measures, innovation measures remain significant. However, we acknowledge that we cannot fully rule out the effect of changes in governance because the law may alter governance along other dimensions that we fail to measure.

Table 9. Pretrends

	(1)	(2)	(3)	(4)	(5)
	<i>R&D/Assets</i>	<i>ln(New class Patent)</i>	<i>ln(New knowledge Patent)</i>	<i>ln(Top 10% Patent)</i>	<i>Patent Value1</i>
Law^{-3}	0.002 (0.001)	−0.024 (0.022)	−0.012 (0.027)	−0.010 (0.014)	−0.002 (0.001)
Law^{-2}	−0.002 (0.002)	0.007 (0.017)	0.016 (0.024)	0.001 (0.021)	−0.002 (0.003)
Law^{-1}	0.002 (0.003)	0.017 (0.012)	0.008 (0.030)	0.008 (0.017)	0.000 (0.003)
Law^0	0.002 (0.002)	−0.006 (0.008)	0.014 (0.027)	−0.001 (0.016)	0.001 (0.003)
Law^{+1}	−0.001 (0.002)	−0.001 (0.017)	0.032 (0.027)	0.014 (0.018)	−0.002 (0.002)
Law^{+2}	0.004* (0.002)	−0.005 (0.016)	−0.010 (0.026)	0.008 (0.016)	0.004* (0.002)
Law^{+3}	0.003** (0.002)	0.037*** (0.013)	0.048*** (0.017)	0.029*** (0.009)	0.005** (0.002)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes
Observations	19,096	19,096	19,096	19,096	19,096
Adjusted R^2	0.666	0.462	0.762	0.680	0.628

Notes. This table presents the effect of pretrends in the difference-in-differences design. We consider five key variables. Law^k is an indicator set to one for year K relative to the adoption of UD laws. Law^{+3} is set to one for years after the third effective year. Control variables, firm, and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level and shown in parentheses.

*10% significance level; **5% significance level; ***1% significance level.

Second, following Fang et al. (2014), we consider additional control variables, including capital expenditure scaled by assets, tangibility, KZ index constructed following Kaplan and Zingales (1997), the Herfindahl–Hirschman Index (HHI), and the squared HHI. The results shown in Table 11 remain robust. Third, the results are robust to standard errors clustered at the incorporated state and year level following

Petersen (2009) as shown in Table 12. Fourth, we use alternative transformations of the patent-based measures as dependent variables. Consistent with the literature, we use the natural logarithm of one plus the raw patent count measures to avoid truncation issues that can arise from zeros in the measures, and we interpret the coefficient as a percentage change. However, it is not precise given the functional form. We thus

Table 10. Robustness Checks: Controlling for Governance Measures

	(1)	(2)	(3)	(4)	(5)
	<i>R&D/Assets</i>	<i>ln(New class Patent)</i>	<i>ln(New Knowledge Patent)</i>	<i>ln(Top 10% Patent)</i>	<i>Patent Value1</i>
<i>Law</i>	0.003*** (0.001)	0.045*** (0.014)	0.087*** (0.022)	0.059*** (0.018)	0.010*** (0.002)
<i>G index</i>	−0.001 (0.000)	0.013 (0.012)	0.016 (0.025)	0.012 (0.018)	0.001 (0.002)
<i>IO</i>	0.005 (0.003)	0.047 (0.030)	0.148*** (0.029)	0.065*** (0.015)	−0.011 (0.007)
Other controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes
Observations	7,734	7,734	7,734	7,734	7,734
Adjusted R^2	0.718	0.459	0.756	0.676	0.653

Notes. This table presents robustness checks of the main results. It considers two corporate governance measures, the *G index* constructed following Gompers et al. (2003), and institutional ownership (*IO*). Baseline control variables, firm, and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level.

***1% significance level.

Table 11. Robustness Checks: Additional Controls

	(1)	(2)	(3)	(4)	(5)
	R&D/Assets	ln(New class Patent)	ln(New Knowledge Patent)	ln(Top 10% Patent)	Patent Value1
<i>Law</i>	0.002** (0.001)	0.020** (0.008)	0.032*** (0.011)	0.019** (0.007)	0.004** (0.002)
Additional controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes
Observations	18,891	18,891	18,891	18,891	18,891
Adjusted R ²	0.676	0.463	0.762	0.681	0.630

Notes. This table presents robustness checks of the main results. It presents the results controlling for additional variables, including the KZ index constructed following Kaplan and Zingales (1997), capital expenditure scaled by assets, tangibility defined as firm's property, plant and equipment scaled by assets, the Herfindah–Hirschman Index defined using assets, and the squared Herfindahl–Hirschman Index. Baseline control variables, firm, and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level.

5% significance level; *1% significance level.

follow previous studies (Burbidge et al. 1988, Levine et al. 2017) and redefine our three patent-based variables using the inverse hyperbolic sine transformation, which addresses the nonnormal distribution of the raw dependent variable. The variable is defined as follows:

$$Var_{i,t} = \text{arcsinh}(Raw_{i,t}) = \ln\left(Raw_{i,t} + \sqrt{Raw_{i,t}^2 + 1}\right), \quad (6)$$

where $Raw_{i,t}$ takes on one of the patent-based measures including *New class Patent*, *New knowledge Patent*, and *Top 10% Patent*. Table 13 confirms that the results remain significant in this alternative definition.

4.7. Heterogeneous Effects

In this section, we extend our analyses by exploring the heterogeneous effects of law adoption. We first examine a firm's exposure to derivative lawsuits. Because it is empirically challenging to gauge the litigation propensity ex ante, we use the realized derivative lawsuits at the industry level to proxy for a firm's exposure to perceived litigation threat. We use the information about derivative suits from Audit Analytics, which covers the period starting

from 2000. The measure for industry litigation risk rests on two assumptions. First, some industries are more prone to lawsuits than others (Francis et al. 1994). The perceived litigation risk in an industry may encourage investors to take legal action, which may not be easily justified in the absence of precedents. Second, the perceived litigation risk persists over time for firms incorporated in states without the UD laws. Because we do not have lawsuit information before 2000, the intensity of shareholder litigation can only be inferred from the following years. We drop all firms incorporated in states that adopted the UD laws because those firms will experience changes in the litigation threat. We first calculate an industry-year-level measure as follows:

$$Industry_Lawsuit_{k,t} = \frac{\sum_{w=1}^W M_{w,t}}{\sum_{v=1}^V M_{v,t}}, \quad (7)$$

where $M_{w,t}$ is the market capitalization of firm w sued by shareholders in year t . W is the total number of sued firms in industry k in year t . V denotes the total number of firms in industry k . We then construct the industry constant measure, $Industry_Lawsuit_k$,

Table 12. Robustness Checks: Two-Way Clustering

	(1)	(2)	(3)	(4)	(5)
	R&D/Assets	ln(New class Patent)	ln(New Knowledge Patent)	ln(Top 10% Patent)	Patent Value1
<i>Law</i>	0.002* (0.001)	0.019* (0.010)	0.032*** (0.011)	0.020** (0.008)	0.004* (0.002)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes
Observations	19,096	19,096	19,096	19,096	19,096
Adjusted R ²	0.667	0.462	0.762	0.680	0.628

Notes. This table presents robustness checks of the main results. It reports the results based on two-way standard errors clustered on the incorporation state and year. Baseline control variables, firm, and headquarter state by year fixed effects are included.

*10% significance level; **5% significance level; ***1% significance level.

Table 13. Robustness Checks: Inverse Hyperbolic Sine Transformation

	(1)	(2)	(3)
	Inverse hyperbolic sine transformation		
	New class Patent	New Knowledge Patent	Top 10% Patent
<i>Law</i>	0.024** (0.010)	0.037** (0.015)	0.025** (0.010)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes
Observations	19,096	19,096	19,096
Adjusted R^2	0.464	0.754	0.674

Notes. This table presents robustness checks of the main results. It presents the results using the inverse hyperbolic sine transformation of the number of patents as dependent variables. Baseline control variables, firm, and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level.

**5% significance level.

which is averaged across years. We split the treatment variable by the median of this measure across industries. *Law* ($\text{Industry_Lawsuit}_k > \text{median}$) is set to one for treated firms in industries with a litigation

risk measure greater than the median in years after law adoption. Conversely, *Law* ($\text{Industry_Lawsuit}_k < \text{median}$) is set to one for treated firms in industries with a litigation risk measure lower than the median.

Table 14. Heterogeneous Effects of UD Laws

	(1)	(2)	(3)	(4)	(5)
	<i>R&D/Assets</i>	$\ln(\text{New class Patent})$	$\ln(\text{New Knowledge Patent})$	$\ln(\text{Top 10\% Patent})$	<i>Patent Value1</i>
Panel A. Subsample: litigation intensity					
<i>Law</i> ($\text{Industry_Lawsuit}_k > \text{median}$)	0.003*** (0.001)	0.025* (0.013)	0.041** (0.016)	0.023** (0.010)	0.004** (0.002)
<i>Law</i> ($\text{Industry_Lawsuit}_k < \text{median}$)	0.000 (0.001)	0.003 (0.020)	0.004 (0.021)	0.008 (0.015)	0.002 (0.003)
Observations	19,096	19,096	19,096	19,096	19,096
Adjusted R^2	0.667	0.462	0.762	0.680	0.628
Panel B. Subsample: institutional ownership					
<i>Law</i> ($IO > \text{median}$)	−0.001 (0.002)	−0.014 (0.017)	0.014 (0.026)	0.012 (0.014)	0.002 (0.002)
<i>Law</i> ($IO < \text{median}$)	0.006** (0.003)	0.057*** (0.016)	0.051** (0.024)	0.028** (0.014)	0.005* (0.003)
Observations	19,096	19,096	19,096	19,096	19,096
Adjusted R^2	0.667	0.463	0.762	0.680	0.628
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Headquarter state-year FE	Yes	Yes	Yes	Yes	Yes

Notes. This table presents the heterogeneous effects of UD laws by industry litigation intensity and firm's institutional ownership. The treatment variable *Law* is split into two separate dummy variables for firms incorporated in states that adopt the law according to the measures of interest. In Panel A, *Law* ($\text{Industry_Lawsuits} > \text{median}$) is set to one for firms in the industries with above-median litigation intensity. *Law* ($\text{Industry_Lawsuits} < \text{median}$) is defined conversely. The information on derivative lawsuits is from Audit Analytics, which covers the period starting from 2000. We thus restrict the defendant firms to those incorporated in states that did not adopt the UD law and assume that the propensity toward lawsuits remains stable over the sample period for those firms. We first calculate an industry by- year measure, which is defined as the total value of firms involving derivative lawsuits in a two-digit Standard Industrial Classification (SIC) industry in a year scaled by the total market value of firms in the given industry in the given year. We then calculate the average measure at the industry level to measure the industry litigation intensity. In Panel B, *Law* ($IO > \text{median}$) is set to one in the years following the law adoption for firms that have institutional ownership greater than the median of the same two-digit SIC industry firms in the year before the law became effective. *Law* ($IO < \text{median}$) is defined conversely. Control variables, firm, and headquarter state by year fixed effects are included. The standard errors are clustered at the incorporation state level and shown in parentheses.

*10% significance level; **5% significance level; ***1% significance level.

As shown in Panel A of Table 14, the positive effects of UD laws on innovation are concentrated among firms in industries with a higher litigation intensity, whereas the coefficients for firms in industries with a low litigation intensity are insignificant.

We next explore the heterogeneous effects by firm's institutional ownership. Our identification is based on the variation in the difficulties of bringing derivative lawsuits for individual investors because of the laws. We thus posit that firms primarily owned by small investors have greater exposure to the law enactment. In contrast, institutional investors can exert significant influence on the firms that they invest in through voting or exits regardless of the adoption of UD laws. In shareholder litigation, institutional investors often proceed with direct suits (class actions) as the leading plaintiffs instead of derivative suits to discipline the management (Cheng et al. 2010). We formally test the heterogeneity of the laws by splitting our baseline treatment variable. *Law* ($IO > median$) is set to one for firms with institutional ownership greater than the median of Compustat firms in the year before the law is effective. *Law* ($IO < median$) is defined as the opposite. The results in Panel B of Table 14 show that the positive effects are concentrated among firms with low institutional ownership.

4.8. Discussion

In this study, we use R&D investment and patents that are eventually granted to proxy for innovation activities, which is consistent with the literature. However, as documented in D'Acunto (2015), nonpatented innovation actually accounts for the majority of the innovation activities within firms. We next discuss how those nonpatented activities affect the interpretation of our findings. First, although we cannot directly gauge those nonpatented activities because of the data availability issue, R&D expenditure will capture some of the related investments. U.S. Generally Accepted Accounting Principles (GAAP) requires that all R&D costs within the scope of Accounting Standards Codification (ASC) 730 are expensed as incurred.²³ ASC 730 covers activities including investigations to gain new knowledge and applications of research findings in the production of new devices, products, and processes. Therefore, the increased R&D may be channeled to innovation activities that do not necessarily generate patents. Second, those nonpatented activities are less observable than patented innovation. Because of the lack of independent certification of originality and relevance, shareholders are more likely to believe that nonpatented innovation is a waste of resources. Managers might be more averse to nonpatented innovation when faced with more external pressure. Shareholder pressure imposed by litigation threat may, therefore, affect nonpatented innovation

to a larger extent than patented innovation. Again, because we do not have data on nonpatented activities, we cannot directly test this prediction.

The nonrandom timing of the law enactment may create concerns for our design. Firms are known to influence lawmaking and enactment through lobbying (see Bertrand et al. 2014 for details). Unfortunately, we cannot precisely identify the lobbying activities related to the universal demand laws because of the limitations of the existing databases. We next assess whether our interpretation of the results will change owing to the reverse causality issue. Theoretically, firms have multiple reasons for lobbying for universal demand laws, which increase the procedural difficulties for shareholders. First, managers who gain private benefits from the agency problems might lobby for the laws to block shareholder litigations. This possibility is in line with the disciplining hypothesis. Because we find that innovation strategies in general benefit shareholders after the law adoption, the empirical results do not support this possibility. Second, firms that plan to engage in radical innovation may be concerned about the potential shareholder lawsuits triggered by failures of their innovation experiments, and thus, they have incentives to lobby for the adoption of the UD laws. Firms may delay investments in innovation until the laws are passed. However, this process is also consistent with the interpretations of our findings because the underlying force for lobbying is managers' recognition of the detrimental effect of shareholder lawsuits on innovative activities. In summary, we investigate whether the shareholder pressure is a valid factor of corporate innovations instead of aiming to establish the causal impact of UD laws on innovation. As a result, the endogenous timing of the laws may not contradict our evidence supportive of the pressure hypothesis.

5. Conclusion

Shareholder litigation as a legal right under corporate law is assumed to help reduce agency costs and to improve corporate governance. However, the external pressure imposed by the threat of shareholder litigation can create distorted managerial incentives, such as short-termism, thereby impeding innovation. In this study, we evaluate the effect of shareholder litigation on corporate innovation, and thus, we attempt to link shareholder protection rights to the real economy.

For identification, we explore the staggered implementation of UD laws across 23 states from 1989 to 2005. The UD laws act as barriers to shareholders filing derivative lawsuits and have significantly reduced the litigation risk *ex ante* for managers in the affected firms. Using a difference-in-differences approach and a matched sample, we find that firms incorporated in

states with UD laws invest more in innovation, engage in more explorative innovation, produce more high-quality patents as measured by citations, and generate a larger patent value. The findings are robust to the inclusion of stringent control variables. Further analyses show that the effect of the UD laws is mainly driven by firms operating in industries with higher levels of litigation and firms with lower levels of institutional ownership.

The evidence suggests that a regulation that controls abusive shareholder litigation can be beneficial to shareholders in terms of encouraging innovation. Our analysis thus extends the discussion of how shareholder intervention shapes corporate policies and contributes to the general policy debate about the merits of shareholder lawsuits.

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Appendix. Variable Definitions

Variable	Definition
<i>Law</i>	An indicator variable equal to one if a firm is incorporated in a state that has universal demand laws in the given year and zero otherwise; source: state statutes/session laws, Appel (2015)
<i>R&D/Assets</i>	R&D expenditure (Compustat variable <i>xrd</i>) scaled by total assets (Compustat variable <i>at</i>); R&D is treated as zero when missing; source: Compustat
<i>ln(Patent)</i>	Natural logarithm of one plus the firm's total number of eventually granted patents filed in a given year; source: NBER
<i>ln(New class Patent)</i>	Natural logarithm of one plus the number of patents filed in technology classes previously unknown to the firm in a given year; source: NBER
<i>ln(Known class Patent)</i>	Natural logarithm of one plus the number of patents filed in a technology class previously known to the firm in a given year; source: NBER
<i>ln(New knowledge Patent)</i>	Natural logarithm of one plus the firm's total number of new knowledge patents filed in a given year; a patent is defined as a new knowledge patent if at least 90% of the backward citations are not from existing knowledge, which includes all of the patents that the firm produced and all of the patents that were cited by the firm's patents filed over the past five years; source: NBER
<i>ln(Known knowledge Patent)</i>	Natural logarithm of one plus the firm's total number of known knowledge patents filed in a given year; a patent is defined as a known knowledge patent if at least 90% of the backward citations are from existing knowledge, which includes all of the patents that the firm produced and all of the patents that were cited by the firm's patents filed over the past five years; source: NBER
<i>ln(Top X% Patent)</i>	Natural logarithm of one plus a firm's total number of patents that are in the top X% category of the distribution of forward citations for patents granted to the U.S. public firms in a given technological class and application year; source: NBER, PatentsView
<i>Patent Value1</i>	Total patent value over the market value of equity; the economic value of patents is constructed following Kogan et al. (2017); source: Center for Research in Security Prices (CRSP), NBER
<i>Patent Value2</i>	Total patent value over the market value of equity; the economic value for each patent is constructed in the spirit of Kogan et al. (2017); when multiple patents are announced on the same date, the value of each patent is calculated using the number of forward citations as weights; source: CRSP, NBER, PatentsView
<i>Patent Value3</i>	Total patent value over the market value of equity; the value for each patent is constructed in the spirit of Kogan et al. (2017); the market reactions are measured using the market model; when multiple patents are announced on the same date, the value of each patent is calculated using the number of forward citations as weights; source: CRSP, NBER, PatentsView
<i>Size</i>	Natural logarithm of total assets (Compustat variable <i>at</i>); source: Compustat
<i>MTB</i>	Firm's market-to-book ratio calculated as (market value of equity (<i>csho</i> × <i>prcc_f</i>) plus book value of assets (<i>at</i>) minus book value of equity (<i>seq</i>) minus balance sheet deferred taxes (<i>txdb</i> ; set to zero if missing) divided by book value of assets (<i>at</i>)); source: Compustat
<i>Leverage</i>	Ratio of long-term liability (Compustat variable <i>lt</i>) net of deferred taxes (Compustat variable <i>txdb</i>) scaled by total assets (Compustat variable <i>at</i>); source: Compustat
<i>ln(age)</i>	Natural logarithm of the number of years since the first year identified in Compustat; source: Compustat

Appendix. (Continued)

Variable	Definition
ROA	Operating income before depreciation (Compustat variable <i>oibdp</i>) scaled by total assets (Compustat variable <i>at</i>); source: Compustat
G index	Index of 24 governance provisions; G index is backfilled to 1981; source: RiskMetrics
IO	The average percentage of shares owned by institutional investors in a given year; source: Thompson Reuters Institutional Holding (13F) Database

Endnotes

¹ Silicon Graphics' chief executive officer (CEO) McCracken testified that shareholder litigation creates an "uncontrolled tax on innovation." His statement was part of a congressional subcommittee hearing on private litigation under the federal securities law (Seligman 1994).

² Other studies emphasize how legal institutions that protect corporate stakeholders, such as creditors and employees, affect innovation (Acharya et al. 2013, 2014). In contrast, we contribute to the literature by focusing on the effect of the shareholder protection laws and specifically, the right of shareholder litigation on innovation.

³ For example, only 10.4% to 15.3% of drug candidates are eventually approved by the U.S. Food and Drug Administration (Hay et al. 2014).

⁴ For example, the stock price of the biotechnology company Alnylam Pharmaceuticals crashed by about 50% after a failed clinical trial. See <http://fortune.com/2016/10/06/alnylam-patient-deaths/>.

⁵ The allegations usually include information-related activities, value-destroying investment decisions, or mismanagement issues.

⁶ Also see the statement from Edward R. McCracken, President of Silicon Graphics: "companies can be exposed to potential litigation whenever the stock price falls by approximately 10%, even if there's absolutely no violation of security laws or fiduciary responsibility."

⁷ From Richard J. Egan, Chairman of EMC Corp. Also see the statement from Thomas Dunlap, Jr., General Counsel of Intel Corp: "Companies will not take sound risks, but will manage their operations so as to maintain steady performance and avoid stock fluctuations" (Seligman 1994, p. 439).

⁸ Agency problems arise in the process of shareholder litigation because the shareholder is acting as the principal and the attorney is acting as the agent. Attorneys might urge shareholders to file lawsuits to maximize their own interests instead of those of shareholders. The problem results in "frivolous" shareholder lawsuits that waste corporate resources.

⁹ We validate this finding empirically in following discussions.

¹⁰ Although our findings are not consistent with the "disciplining hypothesis" in the context of innovation activities, they do not suggest that shareholder litigation cannot effectively mitigate agency problems in other operating dimensions.

¹¹ See He and Tian (2018) for a review.

¹² This study focuses on shareholder litigation rather than the litigation of patents (see the study by Lerner (2010) of patent litigation).

¹³ An example of a Tesla shareholder derivative suit is given in the online appendix.

¹⁴ To increase the probability of winning the suit, shareholders usually allege these misconducts instead of directly accusing firm's innovation-related activities.

¹⁵ Laws regulating shareholder derivative litigation in emerging markets typically resemble the derivative actions in the United States.

India's new Company Bill was introduced by the Ministry of Corporate Affairs and is clear about shareholders' rights to file derivative lawsuits against mismanagement. Shareholder derivative action was first established through regional courts in Shanghai and Jiangsu province and later, written into China's 2005 Company Law.

¹⁶ For example, Lawrence J. Ellison, the CEO of Oracle, agreed to pay \$100 million to charity to settle a derivative lawsuit. He also paid \$22 million to the plaintiffs' counsel in legal fees and expenses related to the case. See http://www.nytimes.com/2005/09/12/technology/oracles-chief-in-agreement-to-settle-insider-trading-lawsuit.html?_r=0.

¹⁷ Legal researchers commonly believe that most derivative lawsuits are meritless and mainly driven by the settlement fees rather than corporate governance issues. The market does not upgrade a firm when the judicial decisions that allow a derivative suit to continue are announced (see Fischel and Bradley 1985, Brandi 1993).

¹⁸ Shareholders can argue futility if the board is believed to be responsible for the wrongdoing and therefore, cannot make unbiased decisions regarding the demand.

¹⁹ The data are available at <http://www3.nd.edu/~mcdonald/10-K-Headers/10-K-Headers.html>.

²⁰ Rule 23 (a) of Federal Rule of Civil Procedure include four requirements for shareholder class actions: (1) the class is so numerous that joinder of all members is impracticable, (2) there are questions of law or fact common to the class, (3) the claims or defenses of the representative parties are typical of the claims or defenses of the class, and (4) the representative parties will fairly and adequately protect the interests of the class. See https://www.law.cornell.edu/rules/frcp/rule_23.

²¹ We consider sample periods from 1994 because the historical incorporation variable from SEC filings starts in 1994. Our results remain unchanged if we use the incorporation state variable combined with the variable from Compustat and extend the sample to 1976.

²² See details of ASC 730 at <https://onlinelibrary.wiley.com/doi/10.1002/9781119575535.ch46>.

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